Practice 8 — Recurrence Relations

Solutions to Selected Problems

Problems 2, 6

Discrete Mathematics — QYSU

Contents

1 Problem 2 — Solving a Second-Order Linear Recurrence **2**

2 Problem 6 — Binary Strings with Two Consecutive Zeros **5**

1 Problem 2 — Solving a Second-Order Linear Recurrence

Theory Recap: Second-Order Linear Recurrences with Constant Coefficients

A **second-order linear recurrence with constant coefficients** has the form

$$a_n = c_1\, a_{n-1} + c_2\, a_{n-2}, \qquad n \geq 2,$$

where c_1, c_2 are constants and a_0, a_1 (or a_1, a_2) are given initial values.

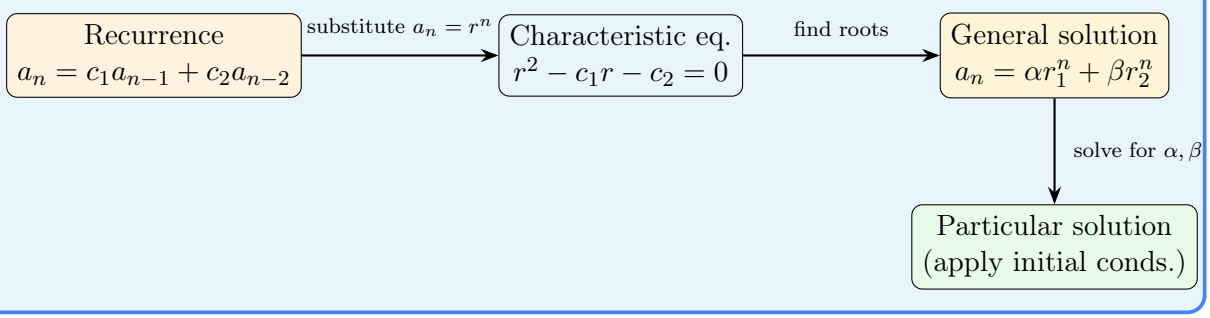
The Characteristic Equation Method. Substitute the trial solution $a_n = r^n$ into the recurrence to obtain the **characteristic equation**:

$$r^2 - c_1\, r - c_2 = 0.$$

The form of the general solution depends on the roots r_1, r_2:

Case	Roots	General Solution
Distinct real roots	$r_1 \neq r_2$, both real	$a_n = \alpha\, r_1^n + \beta\, r_2^n$
Repeated real root	$r_1 = r_2 = r$	$a_n = (\alpha + \beta\, n)\, r^n$
Complex conjugate roots	$r_{1,2} = \rho\, e^{\pm i\theta}$	$a_n = \rho^n \big(\alpha \cos n\theta + \beta \sin n\theta\big)$

The constants α and β are determined by substituting the initial conditions into the general solution and solving the resulting 2×2 linear system.



Problem Statement

Solve the linear recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2}, \qquad n > 2,$$

with initial conditions $a_1 = 2$ and $a_2 = 10$.

Solution

Which case? After finding the roots of the characteristic equation, select the general solution form:

- Distinct real roots $r_1 \neq r_2$: $a_n = \alpha\, r_1^n + \beta\, r_2^n$.
- Repeated real root $r_1 = r_2 = r$: $a_n = (\alpha + \beta\, n)\, r^n$.
- Complex conjugate roots $\rho\, e^{\pm i\theta}$: $a_n = \rho^n(\alpha \cos n\theta + \beta \sin n\theta)$.

Step 1: Write the characteristic equation.

The recurrence $a_n - 5a_{n-1} + 6a_{n-2} = 0$ has characteristic equation

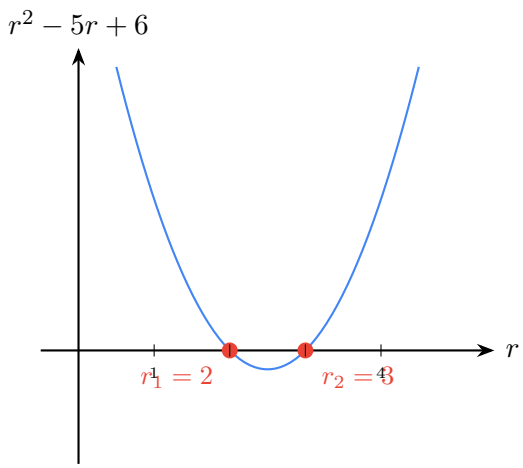
$$r^2 - 5r + 6 = 0.$$

Step 2: Find the roots.

Factoring:

$$r^2 - 5r + 6 = (r-2)(r-3) = 0 \qquad \Longrightarrow \qquad r_1 = 2, \quad r_2 = 3.$$

The roots are distinct and real, so we are in **Case 1**.



Step 3: Write the general solution.

Since $r_1 = 2$ and $r_2 = 3$ are distinct real roots:

$$a_n = \alpha \cdot 2^n + \beta \cdot 3^n.$$

Step 4: Apply initial conditions.

Substituting $n = 1$ and $n = 2$:

$$\begin{cases} a_1 = 2\alpha + 3\beta = 2 \\ a_2 = 4\alpha + 9\beta = 10 \end{cases}$$

Note. This is an ordinary 2×2 linear system—no recurrence tricks needed, just standard algebra.

From the first equation:

$$\alpha = \frac{2 - 3\beta}{2} = 1 - \frac{3\beta}{2}.$$

Substituting into the second equation:

$$4\left(1 - \frac{3\beta}{2}\right) + 9\beta = 10 \quad \Longrightarrow \quad 4 - 6\beta + 9\beta = 10 \quad \Longrightarrow \quad 3\beta = 6 \quad \Longrightarrow \quad \beta = 2.$$

Then $\alpha = 1 - \dfrac{3 \cdot 2}{2} = 1 - 3 = -2$.

Step 5: Write the particular solution.

$$a_n = -2 \cdot 2^n + 2 \cdot 3^n = \boxed{2\big(3^n - 2^n\big)}.$$

Step 6: Verify.

n	$2(3^n - 2^n)$	Expected	Match
1	$2(3-2) = 2$	$a_1 = 2$	✓
2	$2(9-4) = 10$	$a_2 = 10$	✓
3	$2(27-8) = 38$	$5(10) - 6(2) = 38$	✓
4	$2(81-16) = 130$	$5(38) - 6(10) = 130$	✓

$$a_n = 2\big(3^n - 2^n\big)$$

2 Problem 6 — Binary Strings with Two Consecutive Zeros

Theory Recap: Complementary Counting and Recurrences for Constrained Strings

Complementary counting. When counting objects that satisfy a property P, it is sometimes easier to count those that do *not* satisfy P:

$$|P| = |\text{total}| - |\text{not } P|.$$

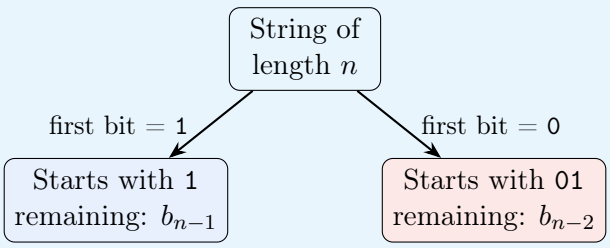
Setting up recurrences for constrained strings. To count binary strings of length n satisfying (or avoiding) a pattern:

1. Define a sequence: let b_n be the count of valid strings of length n.
2. **Condition on the first (or last) character.** Each choice constrains what can follow, yielding a recurrence.
3. Solve using base cases.

Example pattern: avoiding "00". For a string of length n with no two consecutive 0s:

- If it starts with `1`: the remaining $n-1$ characters form any valid string $\longrightarrow b_{n-1}$ choices.
- If it starts with `0`: the next character *must* be `1` (to avoid "00"), and the remaining $n-2$ characters form any valid string $\longrightarrow b_{n-2}$ choices.

This gives the Fibonacci-like recurrence $b_n = b_{n-1} + b_{n-2}$.



Problem Statement

How many binary strings of length 10 contain two consecutive zeros?

Solution

We use **complementary counting**.

Step 1: Count all binary strings of length 10.

Each position has 2 choices, so the total is $2^{10} = 1024$.

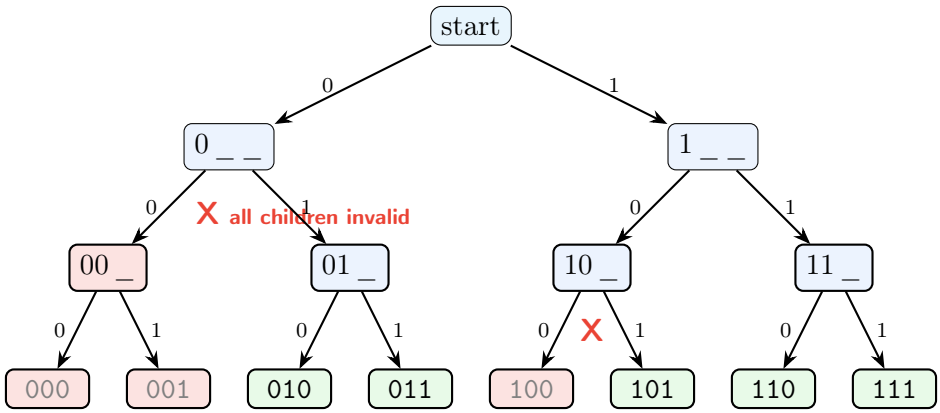
Step 2: Count binary strings of length 10 with *no* two consecutive zeros.

Let b_n denote the number of binary strings of length n with no two consecutive 0s.

Clarification. "Contains two consecutive zeros" means the string has *at least one* occurrence of the substring "00".

Grounding the recurrence. For $n = 3$, the valid strings (no "00") are: `010`, `011`, `101`, `110`, `111` — exactly $5 = b_3$. Check: every string starting with `1` leaves 2 positions ($b_2 = 3$ valid), and every string starting with `01` leaves 1 position ($b_1 = 2$ valid), giving $3 + 2 = 5$. ✓

Decision tree for $n = 3$. The following tree traces every bit choice for a 3-bit string, marking branches that produce "00" as invalid (X). Valid strings (no "00") are shown at the surviving leaves.



Surviving leaves (green): `010`, `011`, `101`, `110`, `111` — exactly $\boxed{b_3 = 5}$, confirming the recurrence $b_3 = b_2 + b_1 = 3 + 2 = 5$.

Deriving the recurrence. Consider a valid string of length n:

- If it starts with `1`: the remaining $n - 1$ characters can be any valid string of length $n - 1$, giving b_{n-1} choices.
- If it starts with `0`: the next character must be `1` (otherwise we have "00"), and the remaining $n-2$ characters can be any valid string of length $n-2$, giving b_{n-2} choices.

Therefore:

$$\boxed{b_n = b_{n-1} + b_{n-2}}, \qquad n \geq 3.$$

Base cases.

- $b_1 = 2$: the strings are `0` and `1`.
- $b_2 = 3$: the valid strings are `01`, `10`, `11` (excluding `00`).

Step 3: Compute b_n for $n = 3, 4, \ldots, 10$.

n	b_n	**Verification** $(b_{n-1} + b_{n-2})$
1	2	(base case: `0`, `1`)
2	3	(base case: `01`, `10`, `11`)
3	5	$b_2 + b_1 = 3 + 2 = 5$ ✓
4	8	$b_3 + b_2 = 5 + 3 = 8$ ✓
5	13	$b_4 + b_3 = 8 + 5 = 13$ ✓
6	21	$b_5 + b_4 = 13 + 8 = 21$ ✓
7	34	$b_6 + b_5 = 21 + 13 = 34$ ✓
8	55	$b_7 + b_6 = 34 + 21 = 55$ ✓
9	89	$b_8 + b_7 = 55 + 34 = 89$ ✓
10	144	$b_9 + b_8 = 89 + 55 = 144$ ✓

Observation. The values $b_n = 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$ are the Fibonacci numbers $F_3, F_4, \ldots, F_{12}$. In general,

$$\boxed{b_n = F_{n+2}}, \qquad \text{where } F_1 = 1,\ F_2 = 1,\ F_3 = 2,\ \ldots$$

Step 4: Apply complementary counting.

$$\text{Strings with “00”} = \text{Total} - \text{Strings without “00”} = 1024 - 144 = \boxed{880}.$$

Verification for small cases. For $n = 2$: total $= 4$, strings without “00” $= 3$ (`01`, `10`, `11`), so strings with “00” $= 1$ (namely `00`). ✓

For $n = 3$: total $= 8$, $b_3 = 5$ (`010`, `011`, `101`, `110`, `111`), so strings with “00” $= 3$ (`001`, `100`, `000`). ✓

The number of binary strings of length 10 that contain two consecutive zeros is $\boxed{880}$.